

Project 2: Exploratory Data Analysis (EDA)

Case File: The Global Bean Coffee Shop — Orders, Customers & Products

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1. Problem Statement

The Global Bean coffee shop maintains three raw data sources — orders.csv, customers.csv, and products.csv — describing what was sold, to whom, and at what price and profit. Management wants to know:

- What does a “typical” order look like (price, size, roast preference)?
- Are there unusual or extreme values (outliers) that distort the average picture?
- Do loyalty-card holders behave differently from non-members?
- Which variables move together (e.g., price vs. profit), and what does that imply for the business?

This report follows the IPO Framework (Input → Process → Output) taught in Project 2: raw evidence is diagnosed, transformed with statistical logic, and translated into a verdict — actionable business insight.

2. Input — The Raw Evidence

Three files were provided:

- orders.csv — 1,000 transactions (coffee type, roast, size, unit price, sales value, loyalty flag)
- customers.csv — 1,000 customer records (name, contact info, address, loyalty flag)
- products.csv — 48 SKUs (coffee type, roast, size, unit price, price/100g, profit)

3. Data Forensics — Diagnosing the Evidence

Before any statistics are calculated, the data's health was checked: data types, missing values, duplicate records, and formatting issues (currency symbols, unit suffixes).

3.1 Orders

orders.csv has zero missing values, but Unit Price and Sales are stored as text (e.g., “\$9.95 ”) and Size includes a unit suffix (e.g., “1.0 kg”). 519 rows are duplicated — but since orders only combine 4 coffee types × 3 roasts × 4 sizes × 2 loyalty flags (at most 96 unique combinations) drawn from 1,000 transactions, repeated rows are an expected characteristic of this dataset, not a sign of data corruption. No rows were dropped for this reason.

3.2 Customers

customers.csv is missing 204 emails (20.4%) and 130 phone numbers (13.0%). Customer IDs themselves are unique (no duplicate records), so this is a genuine completeness gap in contact data. Roughly 1 in 5 customers cannot currently be reached by email.

3.3 Products

products.csv has no missing values and no duplicate rows across its 48 SKUs.

4. Cleaning the Evidence

Unit Price and Sales were converted from currency-formatted strings to numeric floats (stripping the “\$” symbol), and the numeric kg value was extracted from Size. A sanity check confirmed no non-positive or missing values remained after cleaning.

5. Descriptive Statistics — The Five-Number Summary

Min, Q1, Median, Q3, Max — plus mean and count — for the key numeric fields:

Field	Count	Mean	Median	Std Dev	Min	Q1	Q3	Max
Unit Price (\$)	1000	12.91	8.95	9.77	2.69	5.37	15.85	36.46
Sales (\$)	1000	45.14	29.79	42.22	2.69	15.54	59.70	218.73
Size (kg)	1000	1.04	0.50	0.88	0.20	0.50	1.00	2.50

Mean Sales (\$45.14) is notably higher than Median Sales (\$29.79) — a \$15.35 gap signalling a right-skewed distribution (mean > median), meaning the mean alone would overstate the “typical” transaction.

6. Univariate Analysis — The Shape of the Evidence

Histograms reveal whether Sales and Unit Price follow a symmetrical or skewed distribution.

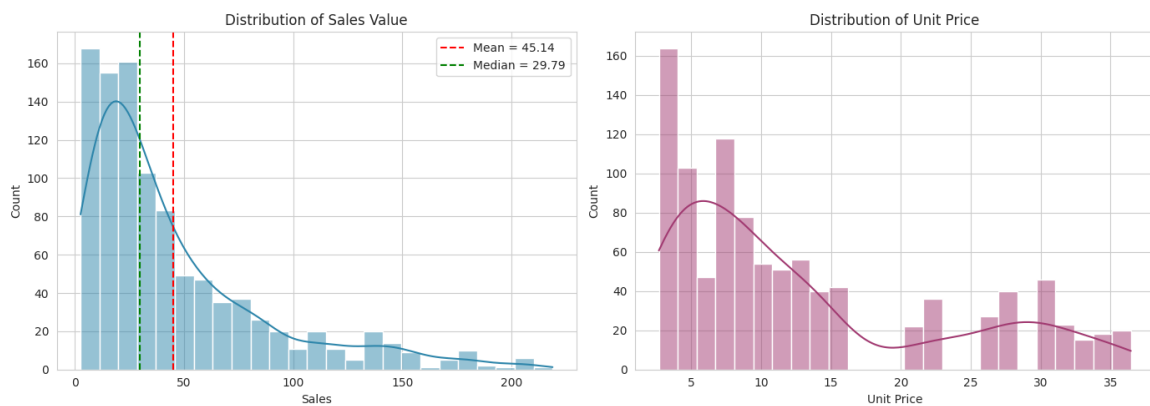


Figure 1. Distribution of Sales Value (left) and Unit Price (right), with mean (red) and median (green) marked.

Sales is right-skewed — most transactions cluster at lower values, with a long tail of high-value orders (large-size + high-unit-price combinations) pulling the mean above the median. The median is the more robust “typical order value” metric to report to stakeholders.

7. Outlier Detection — Unmasking Noise vs. Signal

Two methods were applied, as taught in the training material:

- IQR Method — robust, best for skewed/“dirty” business data. Outlier if $\text{Sales} < Q1 - 1.5 \times \text{IQR}$ or $> Q3 + 1.5 \times \text{IQR}$.
- Z-Score Method — sensitive to extremes, best for roughly normal data. Outlier if $|Z| > 3$.

$Q1 = \$15.54$, $Q3 = \$59.70$, $\text{IQR} = \$44.16 \rightarrow \text{bounds: } [-\$50.70, \$125.94]$.

Method	Outliers Flagged	% of Orders
IQR (1.5×)	74	7.4%
Z-score ($ Z > 3$)	20	2.0%

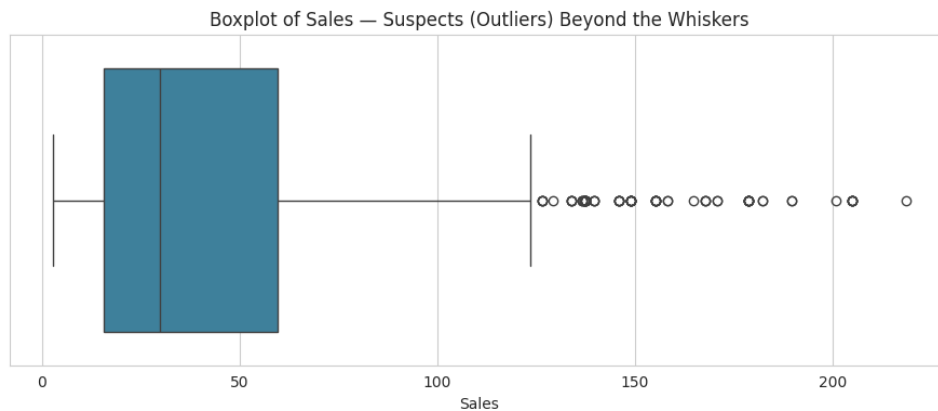


Figure 2. Boxplot of Sales — points beyond the whiskers are the statistically flagged “suspects.”

Diagnosis — Noise vs. Signal: the flagged high-value orders are not data-entry errors (unit price $\times$ implied quantity checks out cleanly for each). They correspond to large-size (2.5 kg), premium purchases. This is Signal, not Noise: a real, legitimate bulk/premium customer segment rather than corrupted data. The correct action is to investigate this segment further (e.g., a bulk/wholesale program) rather than deleting the rows.

8. Trend Analysis — Slicing the Evidence

Average sales were broken down by Coffee Type, Roast Type, Size, and Loyalty Card status.

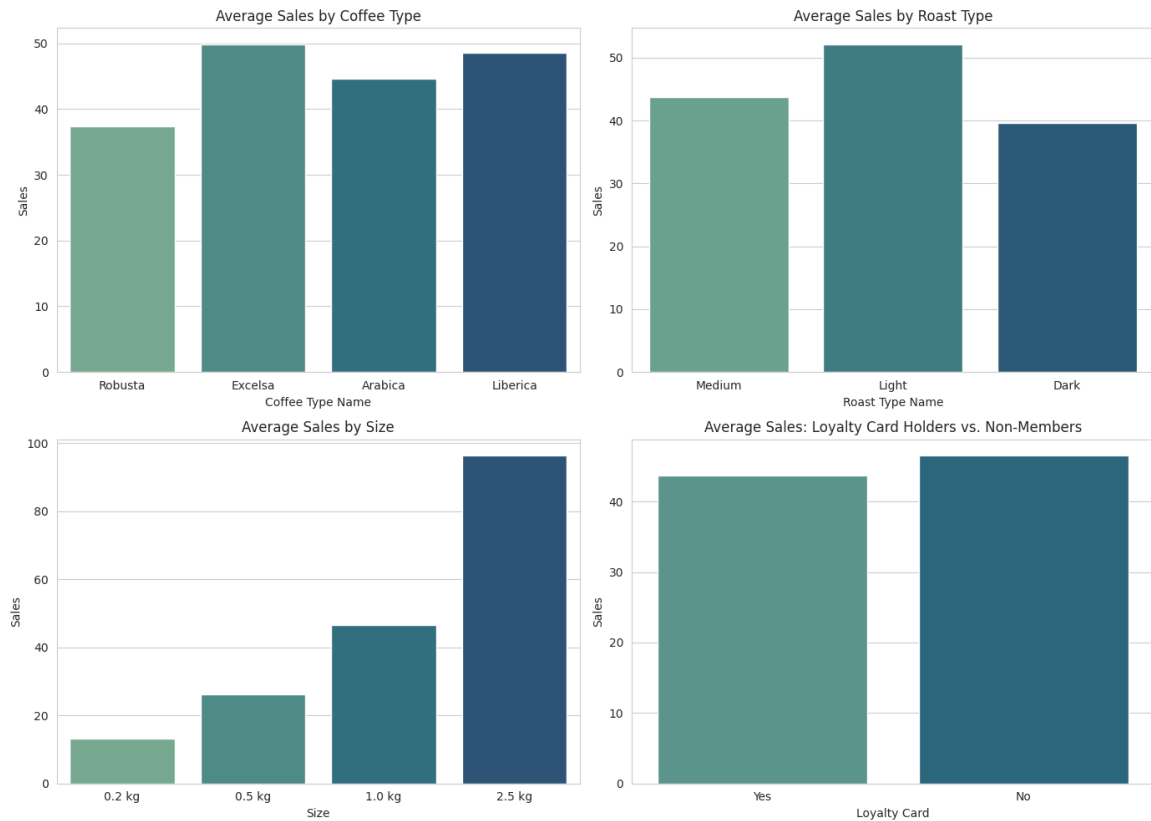


Figure 3. Average Sales by Coffee Type, Roast Type, Size, and Loyalty Card status.

Segment	Category	Mean Sales (\$)
Coffee Type	Robusta	37.37
Coffee Type	Arabica	44.58
Coffee Type	Liberica	48.61
Coffee Type	Excelsa	49.82
Roast Type	Dark	39.58
Roast Type	Medium	43.71
Roast Type	Light	52.12
Size	0.2 kg	13.29
Size	0.5 kg	26.23
Size	1.0 kg	46.66
Size	2.5 kg	96.30
Loyalty Card	Yes	43.67
Loyalty Card	No	46.48

Trend observations: Sales value scales predictably with Size — 2.5 kg bags generate by far the highest average sale. Differences by Coffee Type and Roast Type are more modest, suggesting price is driven mainly by size, not bean variety. Loyalty card holders and non-members spend a similar average amount per transaction (\$43.67 vs. \$46.48) — the loyalty program's value may lie more in purchase frequency than order size, which this transaction-level data can't confirm without a customer-linked order history.

9. Correlation Analysis — Mapping Relationships

Using the Products table (clean numeric fields for every SKU), the Pearson correlation coefficient (r) was computed across Size, Unit Price, Price per 100g, and Profit.

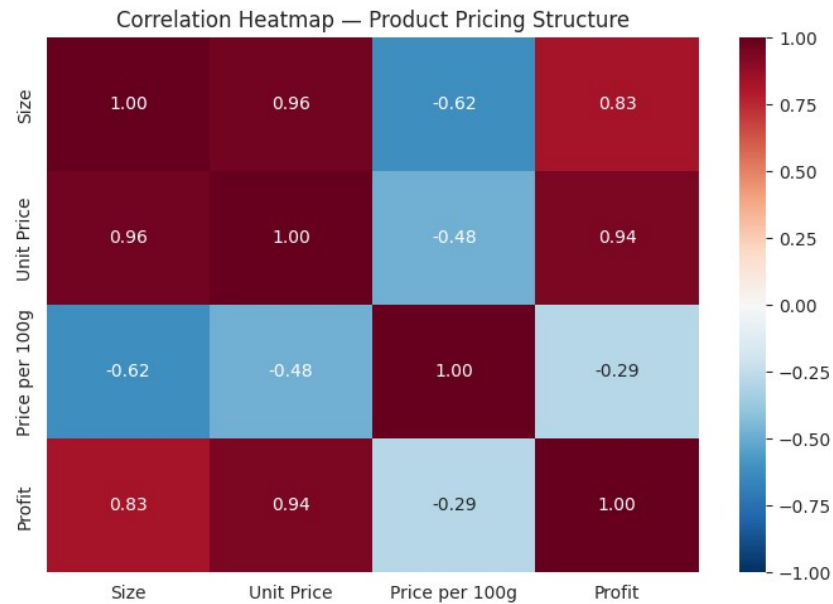


Figure 4. Correlation heatmap of the product pricing structure.

	Size	Unit Price	Price/100g	Profit
Size	1.00	0.96	-0.62	0.83
Unit Price	0.96	1.00	-0.48	0.94
Price/100g	-0.62	-0.48	1.00	-0.29
Profit	0.83	0.94	-0.29	1.00

Interpretation: Size and Unit Price/Profit are strongly positively correlated ($r = 0.96, 0.83$) — bigger bags cost and profit more in absolute terms, which is mechanical and expected. Size and Price per 100g are negatively correlated ($r = -0.62$) — larger bags are cheaper per gram, a classic bulk-discount pricing structure. Golden Rule reminder: these correlations describe the coffee shop's own pricing formula — a designed business rule, not an external causal relationship — a useful distinction before treating any correlation as causal.

10. Customer Base Snapshot

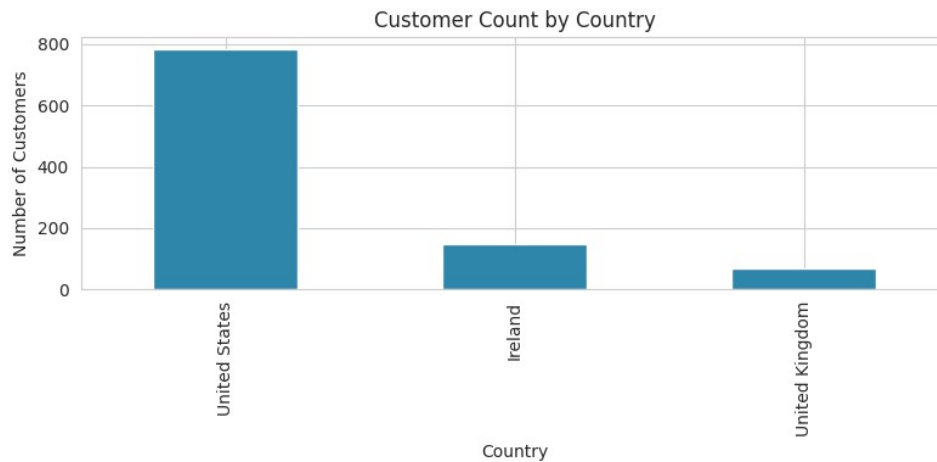


Figure 5. Customer count by country.

Country	Customers	Share
United States	782	78.2%
Ireland	150	15.0%
United Kingdom	68	6.8%

The customer base is heavily concentrated in the United States (78.2%), with Ireland and the UK making up the remainder — useful context for any future segmentation or market-expansion analysis.

11. Synthesizing the Case — The “So What?” Test

What the Data Says	The Business Diagnosis
Sales are right-skewed: mean (\$45.14) > median (\$29.79)	Report the MEDIAN order value to leadership — the mean overstates the “typical” transaction because of a handful of large bulk orders.
7.4% of orders are statistically flagged as outliers, but check out as valid bulk/premium purchases	This is a real customer segment, not bad data — worth a dedicated bulk-buyer/wholesale offer.
20.4% of customer emails and 13.0% of phone numbers are missing	Any email/SMS marketing campaign will miss roughly 1 in 5 customers — prioritize contact-data collection at checkout.
Size is the dominant driver of Sales & Profit; larger bags are cheaper per 100g	Bulk pricing is working as designed — an intentional pricing lever, not a data anomaly.
Customer base is 78.2% US-based	Ireland/UK expansion or region-specific promotions have room to grow before more international marketing spend is justified.

12. Executive Summary

Problem Statement

Understand patterns, trends, distributions, and data quality in The Global Bean coffee shop's orders, customers, and products data.

Methodology

Cleaned currency/unit-formatted fields, checked completeness and duplicates, computed five-number summaries, detected outliers with both IQR and Z-score methods, analyzed trends across coffee type / roast / size / loyalty status, and mapped correlations in the product pricing structure.

Key Findings

- Sales is right-skewed — the median is the more trustworthy “typical order” metric than the mean.
- Statistically flagged high-value outliers are legitimate bulk/premium orders (signal, not noise).
- 20.4% of customers have no email on file and 13.0% have no phone number — a real gap for contact-based marketing.
- Size is the strongest driver of Sales and Profit; the shop's bulk-discount pricing structure (cheaper per-100g at larger sizes) is working as intended.
- The customer base is concentrated in the United States (78.2%), with room to grow in Ireland and the UK.

Recommendations

- Use median (not mean) order value in leadership reporting.
- Investigate a dedicated bulk/wholesale offer for the high-value order segment.
- Prioritize collecting email/phone at checkout to close the contact-data gap.
- Consider region-specific promotions for Ireland/UK before increasing broader international spend.

Verdict: Verified Insight

EDA is the diagnostic phase for data integrity — this analysis translated noise into signal and uncovered the underlying “pulse” of The Global Bean's transactional data, ready to support dashboarding or predictive modeling in the next project milestone.